I. Introduction

A. Background

Automatic speech recognition (ASR) has progressed significantly with deep sequence models. The Transformer architecture [1] and its convolution-augmented variant, the Conformer [2], now achieve state-of-the-art accuracy on benchmarks such as LibriSpeech, outperforming recurrent and CNN models. Self-supervised pretraining [3] further boosts accuracy, and streaming architectures like the RNN-Transducer [4], and its Transformer-based extensions enable low-latency transcription. However, these systems are static: they use a fixed model for all inputs. In realistic deployments, acoustic conditions such as noise, channel and speaker can vary unpredictably over time. A static model may be suboptimal across domains or may waste capacity on easy inputs. A model that performs well under certain acoustic conditions may not generalize equally well to others, leading to fluctuations in transcription quality across inputs. This observation suggests that no single STT system can be expected to perform optimally across all scenarios. Instead, different models often demonstrate strengths in different regimes, forming a collection of complementary experts. Using this diversity requires mechanisms that can adaptively determine which model is most suitable for a given input, rather than committing to a fixed system. Such adaptive strategies aim to retain the efficiency of single-model inference while benefiting from the diversity of multiple pretrained systems.

A key difficulty in this setting is predicting the performance of candidate models without explicitly executing them. Observable properties of the audio signal, such as acoustic statistics, temporal dynamics, and learned representations, provide useful cues for this purpose. Nevertheless, many existing approaches treat each audio segment as an independent unit, making selection decisions based solely on local information. This ignores an important aspect of speech: its inherently sequential nature.

In practical scenarios such as conversations, meetings, or continuous recordings, audio segments are temporally correlated. Consecutive segments often share common attributes, including speaker identity, background conditions, and channel characteristics. Furthermore, transitions between different acoustic regimes typically occur gradually rather than instantaneously. As a result, the suitability of a particular STT model at a given time step is influenced not only by the current segment, but also by its surrounding context.

Neglecting these temporal relationships can lead to unstable or suboptimal selection behavior, particularly in long-form audio where conditions evolve over time. This motivates the development of adaptive STT frameworks that explicitly model sequential dependencies and incorporate contextual information into the selection process. In this work, we explore this direction by formulating STT model selection as a sequential decision problem and introducing a context-aware routing framework that leverages temporal structure to improve selection accuracy.

B. Related Work

Modern end-to-end ASR often uses attention-based encoder-decoder or transducer models. The original Transformer eliminated recurrence in sequence-to-sequence tasks and has been widely adapted for speech. In ASR, the Conformer architecture augments self-attention with convolutional modules to better model local speech structure; it achieved near-state-of-the-art LibriSpeech WERs (2.1. Self-supervised pretraining (e.g. wav2vec 2.0) learns useful speech representations, allowing models to reach very low error rates with limited labels. Streaming extensions include the RNN-Transducer and Transformer-Transducer models (which combine attention with transducer output), as well as chunk-based Conformers for real-time use. These models focus on accuracy and (for streaming ones) low latency, but remain static in that all inputs use the same network.

Streaming ASR places strict latency constraints. RNN-T was a seminal approach for on-the-fly ASR with fixed delay. Recent efforts apply Transformer/Conformer backbones to streaming by attention truncation or recurrence (e.g. Emformer, Streaming Transformer-Transducer) to meet real-time budgets. These approaches optimize architecture to reduce delay but do not adapt computation per input. In contrast, our router is inherently sequential and can be used in a streaming pipeline: at each segment boundary it uses contextual info to pick a model, then immediately invokes that expert (with only its output carrying the latency cost). Thus latency stays low (one model's inference), while adaptation occurs invisibly in parallel.

Routing and Mixture-of-Experts: Our work is inspired by sparse MoE and conditional computation. Early Mixture-of-Experts (Shazeer et al. 2017) enabled enormous capacity by routing tokens to only one expert subnetwork . This idea was massively scaled in Google's GShard (2020) and Follow-up Switch Transformer (2021) : models with hundreds of billions of parameters were trained with sparsely-gated MoE layers at constant cost. GShard demonstrated a 600B-parameter translation model using dynamic routing , while Switch simplified the routing algorithm for efficiency . Beyond NLP, MoE has been applied in vision and other domains as well. All these models perform intra-network routing: each input token or frame is dispatched to expert feedforward layers or networks. By contrast, our router performs inter-model routing: it selects among whole ASR models (or large subnetworks) at each segment. This yields coarse-grained conditional computation: we trade finer token-level routing for more efficient segment-level decisions.

There is growing work on adaptive or multi-domain ASR. You et al. (2021) proposed SpeechMoE, adding MoE layers to a CTC model with a learned embedding for the router . SpeechMoE achieved substantial CER reductions (7–23by dynamically routing frames to experts. A follow-up SpeechMoE2 (2022) improved routing by feeding domain- and accent-embeddings to the router . More recently, Hu et al. (2023) introduced a Switch-Conformer MoE: a multilingual streaming Conformer with MoE FFN layers. They report an 11.9. Our router differs from these in granularity and scope: whereas

SpeechMoE layers route individual frames within a model, our router chooses an entire expert per segment. This yields lower overall switching frequency and allows specialization at the whole-model level.

Conditional Computation and Routing: More broadly, conditional computation in neural networks includes techniques like dynamic skipping of layers (e.g. Skip-RNN, BlockDrop), early-exit classifiers, or adaptive layers. For sequence models, some work (e.g. Routing Transformer) learns sparse attention patterns via content-based routing, reducing complexity from quadratic to sub-quadratic. Roy et al.'s Routing Transformer routes queries to a subset of keys using a k-means router, achieving state-of-the-art language modeling on long contexts . This is complementary to our goal: they reduce attention cost, while we route between entire subnetworks. Other NLP works (e.g. BASELayers) explore sparse capacity allocation for multi-task learning, highlighting issues like expert specialization and training stability. In our speech context, we assume expert models are pre-trained on diverse data, and focus the router's learning on choosing among them. The main trade-off is between granularity and overhead: our segment-level router has fewer switching decisions (favoring stability and efficiency) but may be less responsive to rapid changes than frame-level gating. We mitigate this with temporal context and cost-sensitive loss.

C. Contributions

Our contributions are summarized as follows:

- We introduce a sequential adaptive STT routing framework that extends instance-wise model selection to a sequence-aware setting, where expert selection decisions are conditioned on the temporal context of neighboring audio segments. This enables the system to capture evolving acoustic conditions and improve selection reliability in long-form speech.
- We formulate adaptive STT model selection as a contextual sequence decision problem and propose a transformer-based routing function that maps sequences of segment-level feature representations to expert preference distributions. The model leverages self-attention to capture temporal dependencies such as speaker continuity, environment persistence, and gradual acoustic drift.
- We develop a cost-sensitive training objective based on expected transcription error, which directly optimizes the routing model with respect to task-level performance measures such as WER. The framework naturally supports alternative error metrics, making it loss-agnostic and adaptable to different evaluation criteria.
- We incorporate temporal consistency into the routing process through a context-aware formulation that promotes stable expert selection across consecutive segments, reducing unnecessary switching while preserving responsiveness to meaningful changes in acoustic conditions.
- Our framework performs input-adaptive STT model selection using a lightweight routing mechanism operating on pre-extracted acoustic and embedding-based features. The approach remains feature-agnostic and compatible with arbitrary discriminative representations, while introducing only minimal computational overhead beyond a single STT model execution at inference time.
- The proposed framework is routing-model agnostic and can be instantiated using different sequence modeling architectures. While we focus on transformer-based routing in this work, the formulation allows for alternative temporal models such as recurrent or convolutional sequence learners without modifying the overall framework.

D. Organization

The remainder of the paper is organized as follows. Section II formalizes the sequential adaptive STT selection problem. Section III presents the proposed framework, including the sequential feature representation, transformer-based routing model, and training objective. **??** describes the experimental setup and empirical evaluation. Finally, Section V concludes the paper.

II. Problem Description

We study adaptive speech-to-text (STT) transcription for temporally ordered audio streams composed of a large number of audio segments denoted by $\{x_t\}_{t=1}^{T}$, where T is significantly large. In realistic speech recordings such as conversations, meetings, or long-form audio streams, the acoustic conditions and speaker characteristics evolve over time. Properties of each audio segment, including background noise level, speaker identity, accent, speaking rate, recording quality, and acoustic clarity, may vary gradually or abruptly across the sequence. Consequently, different state-of-the-art STT algorithms may exhibit varying levels of transcription performance depending on the acoustic conditions associated with each segment [5], [6]. Since these conditions change over time and are not known in advance, it is generally unknown which STT algorithm will perform best for each segment in the sequence.

Let $\mathcal{A} = \{A_1, A_2, \ldots, A_K\}$ denote a finite set of K distinct STT algorithms, where $A_j \in \mathcal{A}$ represents the j-th algorithm. For each audio segment x_t, the transcription produced by the j-th STT algorithm is denoted by

$$T_{t,j} = A_j(x_t).$$

Each transcription $T_{t,j}$ can be evaluated using a transcription error measure defined with respect to the ground-truth transcription R_t associated with segment x_t. Using standard evaluation metrics such as the Word Error Rate (WER) or other suitable transcription accuracy measures [6], we define the transcription loss incurred by using algorithm A_j on segment x_t as

$$\ell(T_{t,j}, R_t),$$

where $\ell(\cdot,\cdot)$ denotes the chosen error metric.

In practice, applying all K STT algorithms to every incoming segment in order to select the best transcription is computationally expensive and inefficient. Instead, the goal is to select a single STT algorithm for each segment in a manner that minimizes the overall transcription error across the entire

sequence. Let $j_t \in \{1, 2, \ldots, K\}$ denote the index of the STT algorithm selected for segment x_t. The resulting transcription for that segment is therefore given by

$$\hat{T}_t = A_{j_t}(x_t).$$

The objective is to determine a sequence of algorithm selections $\{j_t\}_{t=1}^{T}$ that minimizes the cumulative transcription error over the entire audio stream:

$$\sum_{t=1}^{T} \ell(T_{t,j_t}, R_t).$$

Unlike independent algorithm selection problems where each decision is made solely based on the current input, the performance of STT algorithms may exhibit temporal correlations across consecutive audio segments due to evolving acoustic environments, speaker continuity, or recording conditions. Therefore, selecting the most suitable algorithm for a given segment may benefit from considering information from neighboring segments within the sequence. In the following section, we introduce a framework that models this sequential algorithm selection problem and determines the STT algorithm for each segment while accounting for temporal dependencies across the audio stream.

III. Workdone

A. Algorithm Overview

The proposed framework extends adaptive STT model selection to the sequential setting by exploiting temporal dependencies across consecutive audio segments. The overall system operates in two phases: an offline training phase and an online inference phase, as illustrated in Fig. **??**. The offline phase learns a routing model that captures how the relative performance of different STT experts evolves across a sequence of audio segments, while the online phase uses the trained routing model to select an appropriate expert for each incoming segment in real time.

In the offline training phase, the system is provided with a collection of labeled audio recordings that are partitioned into temporally ordered segments $\{x_t\}_{t=1}^{T}$. For each segment, a feature representation $\mathbf{z}_t$ is extracted using the feature pipeline described in Section III-B. These representations may include acoustic descriptors, signal statistics, and neural speech embeddings that characterize the acoustic properties of the input segment. In contrast to independent instance-level selection, the framework processes sequences of segment-level representations to capture contextual relationships across neighboring segments. For each training sequence, all STT experts $\{A_1, \ldots, A_K\}$ are evaluated on the corresponding audio segments to obtain their transcription outputs and the associated transcription losses defined in Section II. These evaluations provide supervision signals indicating the relative suitability of each expert for different segments within the sequence.

Using these sequence-level observations, the routing model learns to predict the relative performance of the candidate STT experts conditioned on the contextual representation of the input sequence. The parameters of the routing model are optimized during the offline phase by minimizing the cumulative transcription loss over the training data, allowing the system to learn how temporal patterns in the audio stream relate to the performance of different experts.

In the online inference phase, the system receives a stream of incoming audio segments $\{x_t\}$. For each segment, the feature vector $\mathbf{z}_t$ is extracted as shown in Fig. **??**. The routing model then processes the current segment representation together with its recent temporal context to estimate a weight vector $\mathbf{w}_t$ over the candidate experts. A single STT algorithm A_{j_t} is selected according to the highest predicted weight, and the final transcription $\hat{T}_t$ is obtained by executing only the selected expert on the input segment. Because only one STT model is executed at inference time, the proposed framework retains the computational efficiency of single-model inference while dynamically adapting the expert selection policy to the evolving acoustic conditions of the audio stream.

The detailed components of the proposed framework, including the sequence representation, the routing model, and the training procedure, are described in the following subsections.

Remark 1.

Algorithm 1 Online Sequential STT Model Selection

Inputs:

- Incoming audio segment x_t
- Context window length L
- Set of audio feature extraction functions $\mathbb{F}(\cdot)$
- Trained sequential routing function $g_\theta(\cdot)$
- STT expert pool $\mathcal{A} = \{A_1, \ldots, A_K\}$

Outputs:

- Final transcription $\widehat{T}_t$ for audio segment x_t

Inference Steps:

1: Extract segment-level features:

$$\mathbf{z}_t \leftarrow \mathbb{F}(x_t)$$

2: Update contextual feature sequence:

$$Z_t \leftarrow (\mathbf{z}_{t-L+1}, \ldots, \mathbf{z}_t)$$

3: Compute expert selection scores using the routing model:

$$\mathbf{s}_t \leftarrow g_\theta(Z_t)$$

4: Normalize scores using softmax to obtain routing weights $\mathbf{w}_t$

5: Select expert model:

$$j_t \leftarrow \arg\max_{j \in \{1,\ldots,K\}} w_{t,j}$$

6: Obtain final transcription:

$$\widehat{T}_t \leftarrow A_{j_t}(x_t)$$

7: Return $\widehat{T}_t$

B. Sequential Feature Representation

For each audio segment x_t in the input stream, we construct a compact feature vector $\mathbf{z}_t \in \mathbb{R}^d$ that captures acoustic and contextual characteristics relevant to STT performance, as illustrated in Fig. **??**. These features summarize signal quality, phonetic content, speaker characteristics, and temporal speech dynamics [7], [8].

Given a sequence of audio segments $\{x_t\}_{t=1}^{T}$, the feature extraction process produces a corresponding sequence of feature vectors

$$Z = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T),$$

which serves as a compact representation of the evolving acoustic conditions across the audio stream.

1) Spectral–Phonetic Features: We extract 13 Mel-frequency cepstral coefficients (MFCCs) together with their first- and second-order derivatives (Δ and $\Delta\Delta$) [9], [10]. Lower-order coefficients represent the coarse spectral envelope, while higher-order terms encode finer phonetic detail. The inclusion of temporal derivatives improves robustness to coarticulation effects, speaker articulation differences, and prosodic variation [7]. These descriptors provide a compact summary of the phonetic and spectral characteristics of each audio segment.

2) Prosodic Features: Prosodic features, including pitch statistics (such as mean and range) and silence ratios derived from voice activity detection (VAD), characterize speaking rhythm, stress patterns, and expressiveness [11], [12]. Such descriptors help identify speaking styles or acoustic conditions under which transcription accuracy may degrade, particularly in expressive or noisy speech scenarios [8].

3) Neural Speech Embeddings: High-level speech representations are extracted from pretrained self-supervised encoders such as Wav2Vec2 and HuBERT [3], [13]. These embeddings encode rich information about phonetic identity, speaker characteristics, and semantic context, complementing conventional low-level acoustic features. Since these embeddings can be obtained through a single forward pass of the pretrained encoder without performing full speech decoding, they introduce minimal computational overhead relative to running the STT models themselves.

4) Signal and Temporal Features: Signal-level features including signal-to-noise ratio (SNR), zero-crossing rate (ZCR), and spectral entropy quantify acoustic clarity and spectral variability [14], [15]. Temporal descriptors such as segment duration, speech rate, and pause ratio capture aspects of speaking fluency and pacing, which can significantly influence transcription robustness [16], [17].

The resulting feature vectors $\{\mathbf{z}_t\}_{t=1}^{T}$ collectively provide a compact sequence-level representation of the evolving acoustic conditions within the audio stream. This representation forms the basis for modeling temporal dependencies across segments and serves as the input to the routing mechanism that determines the most suitable STT expert for each segment.

Remark 2.

C. Transformer-Based Routing Model

Given the sequence of segment-level feature vectors

$$Z = (\mathbf{z}_1, \mathbf{z}_2, \ldots, \mathbf{z}_T), \qquad \mathbf{z}_t \in \mathbb{R}^d,$$

obtained from the feature extraction pipeline described in Section III-B, the goal of the routing model is to predict, for each segment x_t, the relative suitability of the candidate STT experts $\mathcal{A} = \{A_1, \ldots, A_K\}$ while accounting for contextual dependencies across neighboring segments.

To achieve this, we define a sequential routing function

$$g_\theta(Z_t) = \mathbf{s}_t \in \mathbb{R}^K,$$

where θ denotes the trainable parameters of the routing model, $Z_t = (\mathbf{z}_{t-L+1}, \ldots, \mathbf{z}_t)$ represents the contextual feature sequence within a temporal window of length L, and $\mathbf{s}_t = (s_{t,1}, \ldots, s_{t,K})$ denotes the routing logits associated with the candidate experts. Each component $s_{t,j}$ represents the predicted preference score indicating how suitable expert A_j is expected to be for transcribing the current audio segment x_t given the contextual acoustic information contained in Z_t.

In our framework, the routing function $g_\theta(\cdot)$ is implemented using a transformer-based sequence model that maps the input feature sequence Z_t into a contextual representation. Let

$$P_t = (\mathbf{p}_{t-L+1}, \ldots, \mathbf{p}_t)$$

denote the projected feature sequence obtained by applying a linear embedding transformation to each segment-level feature vector

$$\mathbf{p}_i = W_e \mathbf{z}_i + b_e,$$

where $W_e \in \mathbb{R}^{d' \times d}$ and $b_e \in \mathbb{R}^{d'}$ are learnable projection parameters. These embeddings are augmented with positional encodings to preserve the temporal ordering of the segments within the context window.

The embedded sequence P_t is then processed by a stack of transformer encoder layers. Each layer performs multi-head self-attention followed by a position-wise feedforward transformation. For an input sequence representation $H^{(\ell)}$, the attention operation at layer ℓ is defined as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V,$$

where Q, K, and V denote the query, key, and value matrices derived from the input representation $H^{(\ell)}$. Through this self-attention mechanism, the model learns contextual relationships between neighboring segments, enabling the routing function to capture temporal dependencies such as persistent acoustic environments, speaker continuity, or gradual changes in signal conditions.

Let

$$H_t = (\mathbf{h}_{t-L+1}, \ldots, \mathbf{h}_t)$$

denote the contextual representations produced by the final transformer layer. The representation corresponding to the most recent segment, $\mathbf{h}_t$, summarizes the relevant contextual information contained within the temporal window. This representation is used to compute the routing logits through a linear prediction layer

$$\mathbf{s}_t = W_o \mathbf{h}_t + b_o,$$

where $W_o \in \mathbb{R}^{K \times d'}$ and $b_o \in \mathbb{R}^K$ are learnable parameters.

To obtain normalized routing probabilities over the candidate experts, we apply a softmax transformation to the logits:

$$w_{t,j} = \frac{\exp(s_{t,j})}{\sum_{m=1}^{K} \exp(s_{t,m})}, \qquad j = 1, \ldots, K, \tag{1}$$

where $w_{t,j}$ denotes the predicted probability that expert A_j will produce the most accurate transcription for the current audio segment x_t given its temporal context. These probabilities quantify the model's confidence regarding the relative suitability of the candidate experts.

The routing decision is then obtained by selecting the expert with the highest predicted probability, as summarized in Algorithm 1. The training procedure for estimating the parameters θ of the routing model is described in the following subsection.

Remark 3.

D. Training Objective

The parameters of the sequential routing model are learned in an offline supervised setting using labeled speech data. Let

$$\mathcal{D} = \{(x_1, R_1), \ldots, (x_T, R_T)\}$$

denote a temporally ordered dataset of T audio segments together with their corresponding reference transcriptions. Let $\mathcal{A} = \{A_1, \ldots, A_K\}$ denote the pool of K pretrained STT experts, and let $\mathbb{F}(\cdot)$ represent the feature extraction pipeline described in Section III-B. For each audio segment x_t, the feature extractor produces a segment-level feature vector

$$\mathbf{z}_t = \mathbb{F}(x_t).$$

During training, all experts in $\mathcal{A}$ are evaluated on each audio segment in order to obtain their candidate transcriptions

$$T_{t,j} = A_j(x_t), \qquad j = 1, \ldots, K.$$

These transcriptions are compared with the ground-truth reference transcription R_t to compute the corresponding transcription errors. Using the Word Error Rate (WER) metric, the transcription error associated with expert A_j on segment x_t is defined as

$$\mathrm{WER}(T_{t,j}, R_t) = \frac{S + I + D}{|R_t|},$$

where S, I, and D denote the numbers of substitutions, insertions, and deletions obtained through a standard Levenshtein alignment [6], and $|R_t|$ denotes the number of words in the reference transcription.

Given the contextual feature sequence

$$Z_t = (\mathbf{z}_{t-L+1}, \ldots, \mathbf{z}_t),$$

the transformer routing model described in Section III-C produces routing logits

$$\mathbf{s}_t = g_\theta(Z_t) \in \mathbb{R}^K.$$

These logits are converted into routing probabilities using the softmax function defined in (1), producing the expert weight vector

$$\mathbf{w}_t = (w_{t,1}, \ldots, w_{t,K}),$$

where $w_{t,j}$ represents the predicted probability that expert A_j will yield the most accurate transcription for segment x_t given its contextual acoustic information.

Using these routing probabilities, we define the expected transcription loss for segment x_t as

$$\mathcal{L}_t = \sum_{j=1}^{K} w_{t,j} \, \mathrm{WER}(T_{t,j}, R_t). \tag{2}$$

Intuitively, this formulation encourages the routing model to assign higher probability mass to experts that achieve lower transcription errors on the corresponding audio segment. By minimizing this expected loss, the model learns to predict expert preferences that align with the empirical performance of the STT experts.

Although WER is used as the primary optimization objective due to its widespread adoption and interpretability in speech recognition research [5], [6], other transcription quality metrics such as Character Error Rate (CER) or task-specific evaluation measures could be incorporated into the same training framework without modifying the routing architecture.

The overall training objective minimizes the expected transcription error across the entire dataset:

$$\min_\theta \mathcal{L} = \frac{1}{T} \sum_{t=1}^{T} \mathcal{L}_t = \frac{1}{T} \sum_{t=1}^{T} \sum_{j=1}^{K} w_{t,j} \, \mathrm{WER}(T_{t,j}, R_t). \tag{3}$$

During optimization, the WER values $\mathrm{WER}(T_{t,j}, R_t)$ are treated as fixed constants obtained from offline evaluation of the STT experts. The parameters θ of the transformer routing model are then optimized using gradient-based learning to minimize the expected transcription loss defined in (3). The complete offline training procedure is summarized in Algorithm 2.

Remark 4.

E. Temporal Consistency Regularization

In sequential audio streams, acoustic conditions often evolve gradually over time, and the optimal STT expert may remain stable across consecutive segments. To encourage temporally consistent routing decisions, we introduce a regularization term that penalizes abrupt changes in the predicted expert distributions.

Let $\mathbf{w}_t$ and $\mathbf{w}_{t-1}$ denote the routing probability vectors for consecutive segments. We define the temporal consistency regularization term as

$$\mathcal{L}_{\text{smooth}} = \frac{1}{T-1} \sum_{t=2}^{T} \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_2^2.$$

The overall training objective then becomes

$$\min_\theta \; \mathcal{L}_{\text{total}} = \mathcal{L} + \lambda \mathcal{L}_{\text{smooth}}, \tag{4}$$

where $\lambda \geq 0$ controls the trade-off between transcription accuracy and temporal stability of routing decisions.

This regularization encourages the routing model to produce smoother expert selection trajectories, reducing unnecessary switching between experts while still adapting to meaningful changes in acoustic conditions.

Algorithm 2 Offline Training Procedure for the Sequential Routing Model

Inputs

- $\{(x_1, R_1), \ldots, (x_T, R_T)\}$: Labeled dataset of audio segments and reference transcriptions
- $\mathbb{F}$: Feature extraction pipeline
- $\mathcal{A} = \{A_1, \ldots, A_K\}$: Pool of pretrained STT experts
- Context window length L

Output

Trained routing model g_θ

Training Steps

```
Split dataset into training and validation sets: D_train, D_val
for t ← 1 to |D_train| do
    Extract features: z_t ← F(x_t)
    for each expert A_j ∈ A do
        Transcribe segment: T_{t,j} ← A_j(x_t)
        Compute transcription error: WER_{t,j} ← WER(T_{t,j}, R_t)
    end for
end for
for each training step do
    Construct contextual feature sequence: Z_t = (z_{t−L+1}, ..., z_t)
    Compute routing logits: s_t ← g_θ(Z_t)
    Compute routing probabilities using softmax
    Compute expected loss L_t using (2)
end for
Optimize model parameters θ using gradient-based learning (e.g., Adam)
Perform early stopping and hyperparameter tuning using validation loss on D_val
```

IV. Simulations

V. Conclusion

REFERENCES

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. 31st Conf. Neural Inf. Process. Syst. (NeurIPS)*, pp. 6000–6010, 2017. [Online]. Available: https://papers.nips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

[2] A. Gulati, J. Qin, C.-C. Chiu, N. Parmar, Y. Zhang, J. Yu, W. Han, S. Wang, Z. Zhang, Y. Wu, and et al., "Conformer: Convolution-augmented transformer for speech recognition," *arXiv preprint arXiv:2005.08100*, 2020. [Online]. Available: https://arxiv.org/abs/2005.08100.

[3] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Proc. 34th Conf. Neural Inf. Process. Syst. (NeurIPS)*, 2020. [Online]. Available: https://papers.nips.cc/paper/2020/hash/92d1e1eb1cd6f9fba3227870bb6d7f07-Abstract.html.

[4] A. Graves, "Sequence transduction with recurrent neural networks," *ArXiv*, vol. abs/1211.3711, 2012.

[5] D. Yu and L. Deng, "Automatic speech recognition: A deep learning approach," *Springer London*, 2015. Chapters 1–2 discuss applications in healthcare, education, and assistive tech.

[6] J. Li, L. Deng, Y. Gong, and R. Haeb-Umbach, "An overview of noise-robust automatic speech recognition," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 22, no. 4, pp. 745–777, 2014.

[7] L. Rabiner and B.-H. Juang, *Fundamentals of speech recognition*. USA: Prentice-Hall, Inc., 1993.

[8] Y. Li, Z. Zhao, O. Klejch, P. Bell, and C. Lai, "Asr and emotional speech: A word-level investigation of the mutual impact of speech and emotion recognition," *arXiv preprint arXiv:2305.16065*, 2023. [Online]. Available: https://arxiv.org/abs/2305.16065.

[9] S. Davis and P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 28, no. 4, pp. 357–366, 1980.

[10] S. Furui, "Speaker-independent isolated word recognition using dynamic features of speech spectrum," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 34, no. 1, pp. 52–59, 1986.

[11] P. Boersma and D. Weenink, "Praat, a system for doing phonetics by computer," *Glot international*, vol. 5, pp. 341–345, 01 2001.

[12] J. Sohn, N. S. Kim, and W. Sung, "A statistical model-based voice activity detection," *IEEE Signal Processing Letters*, vol. 6, no. 1, pp. 1–3, 1999.

[13] W.-N. Hsu, B. Bolte, Y.-H. H. Tsai, K. Lakhotia, R. Salakhutdinov, and A. Mohamed, "Hubert: Self-supervised speech representation learning by masked prediction of hidden units," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, p. 3451–3460, Oct. 2021.

[14] B. Schuller and A. Batliner, *Computational Paralinguistics: Emotion, Affect and Personality in Speech and Language Processing*. Wiley Publishing, 1st ed., 2013.

[15] F. Eyben, K. R. Scherer, B. W. Schuller, J. Sundberg, E. André, C. Busso, L. Y. Devillers, J. Epps, P. Laukka, S. S. Narayanan, and K. P. Truong, "The geneva minimalistic acoustic parameter set (gemaps) for voice research and affective computing," *IEEE Transactions on Affective Computing*, vol. 7, no. 2, pp. 190–202, 2016.

[16] T. Ko, V. Peddinti, D. Povey, and S. Khudanpur, "Audio augmentation for speech recognition," in *Proceedings of Interspeech 2015*, pp. 3586–3589, Sept. 2015.

[17] Y. Zhang, T. N. Sainath, R. J. Weiss, and A. Narayanan, "Transformer transducer: One model unifying streaming and non-streaming speech recognition," *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 1599–1613, 2020.